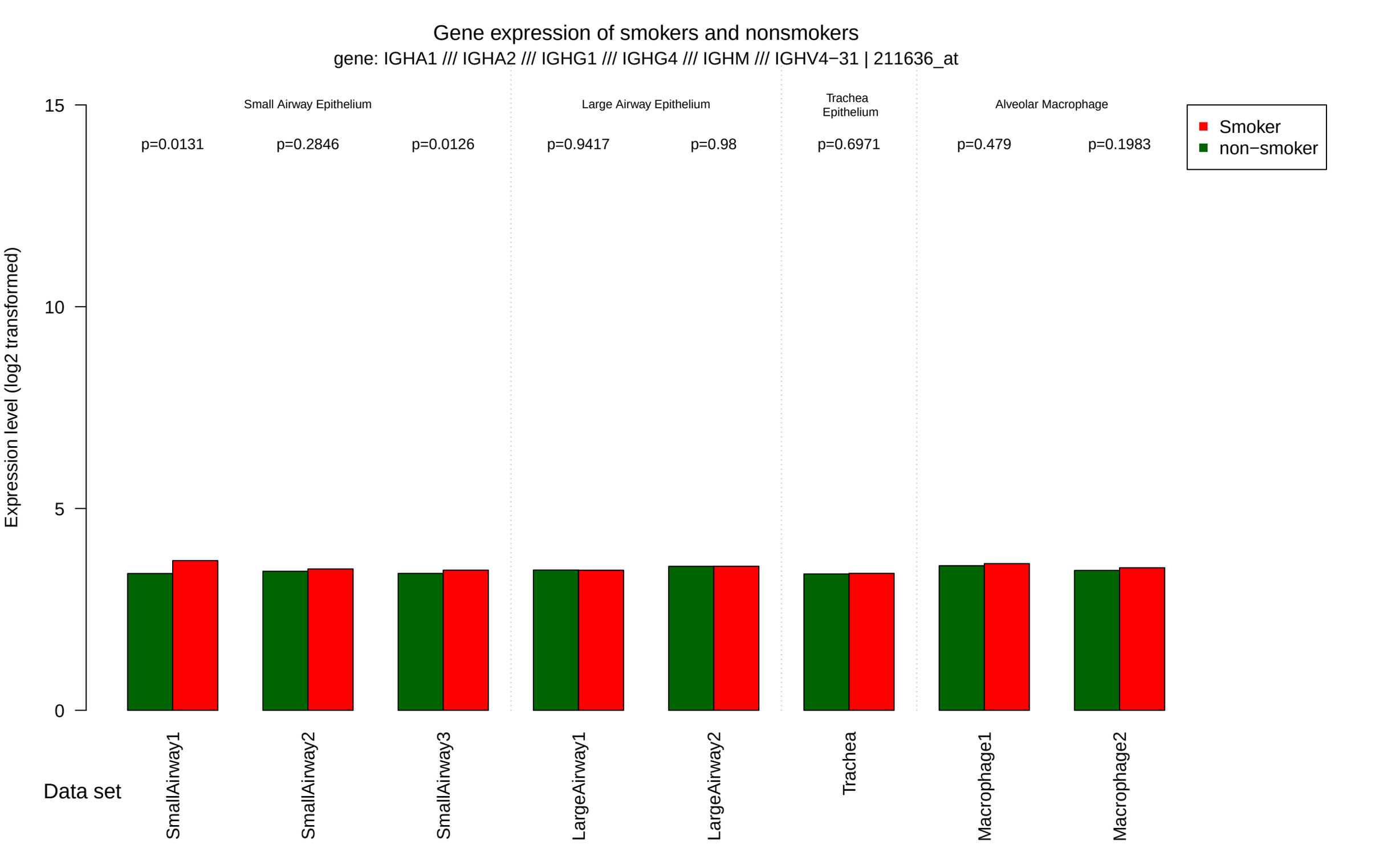
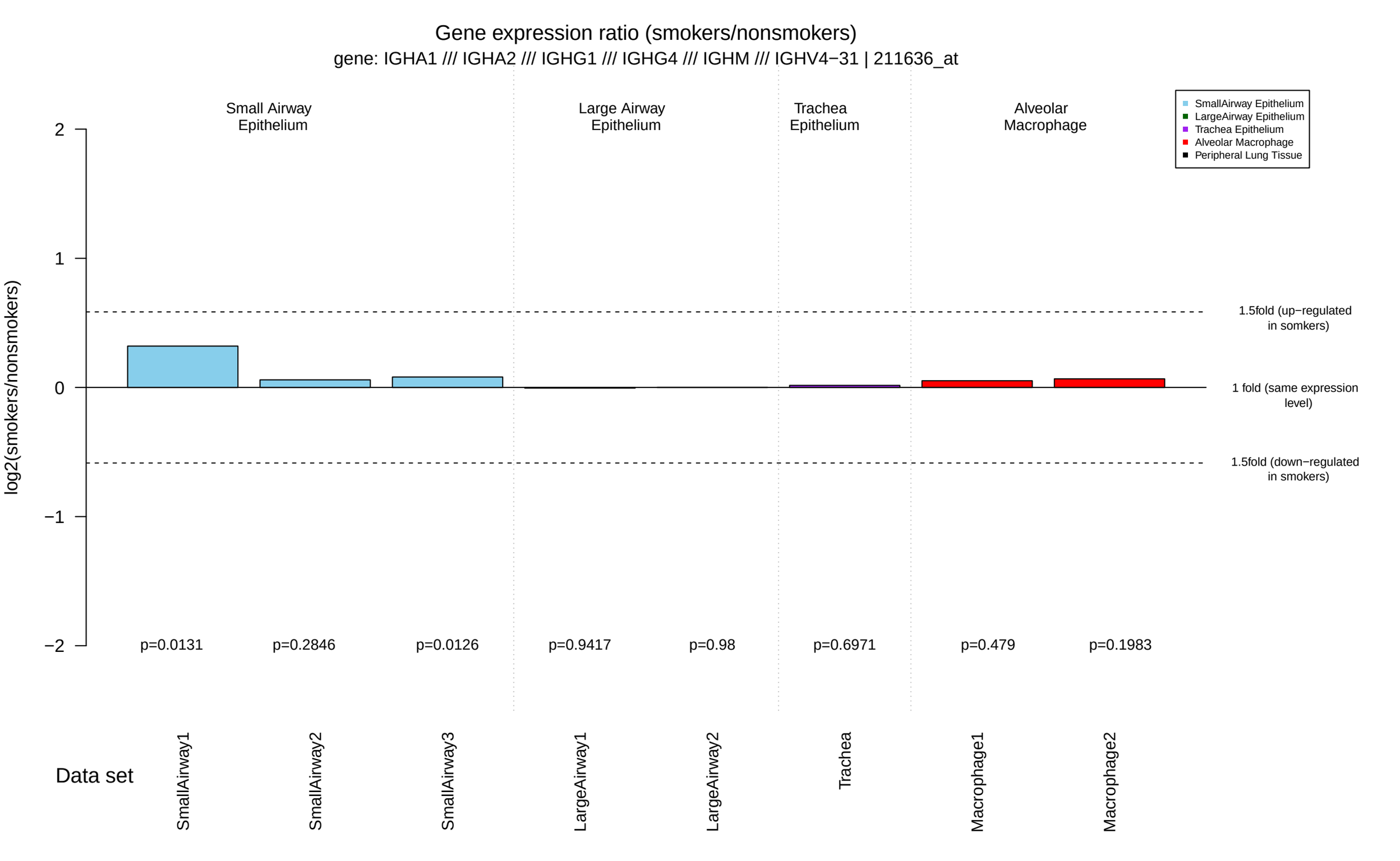
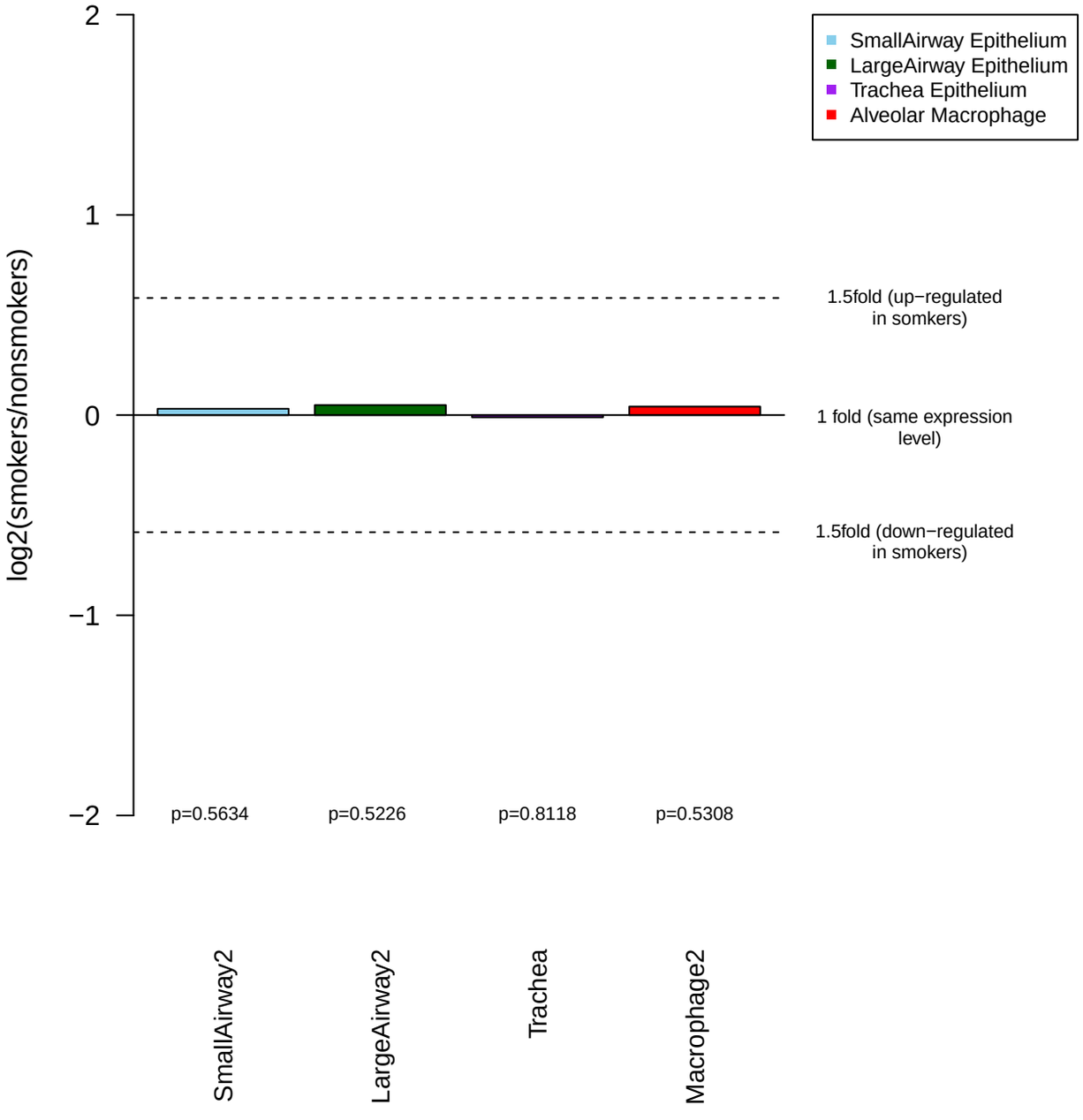

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 211636_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0131
p=0.2846
p=0.0126
p=0.9417
p=0.98
p=0.6971
p=0.479
p=0.1983
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 211636_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0131
p=0.2846
p=0.0126
p=0.9417
p=0.98
p=0.6971
p=0.479
p=0.1983
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

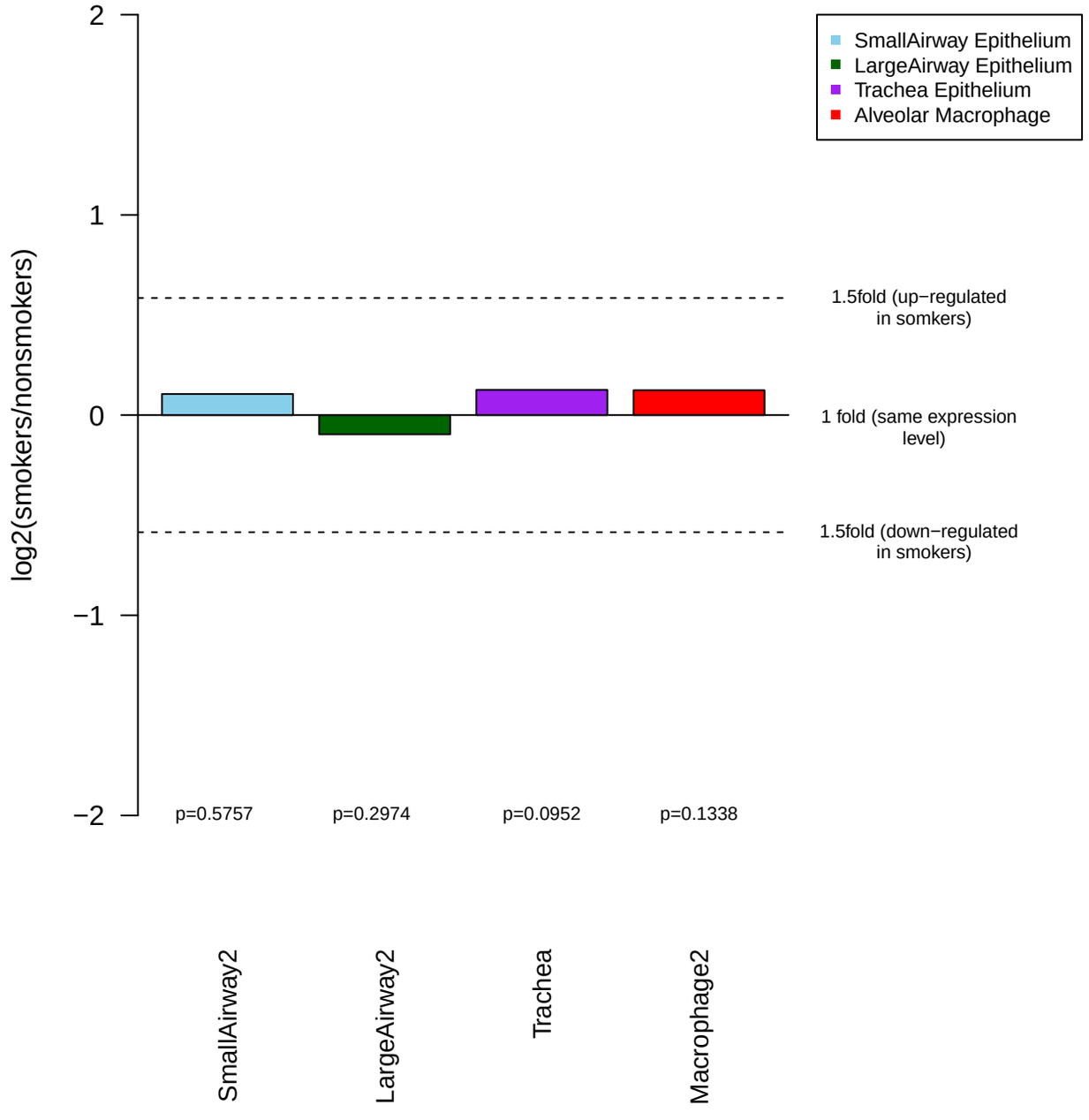
Gene expression ratio in Males

ene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 211636_at

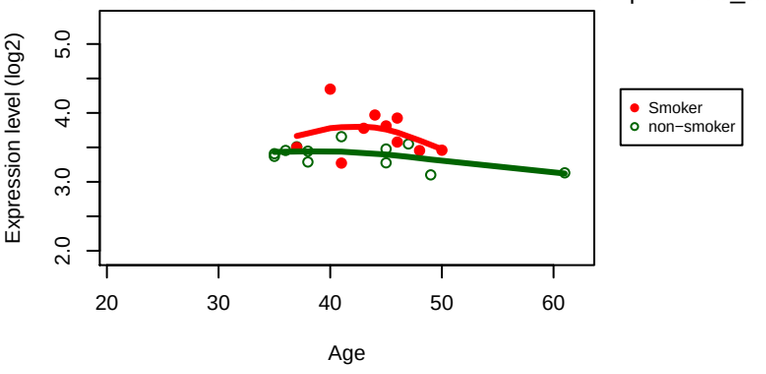


Gene expression ratio in Females

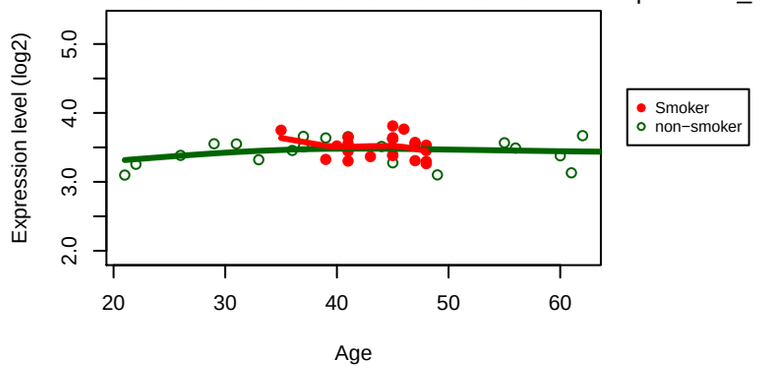
gene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 211636_at



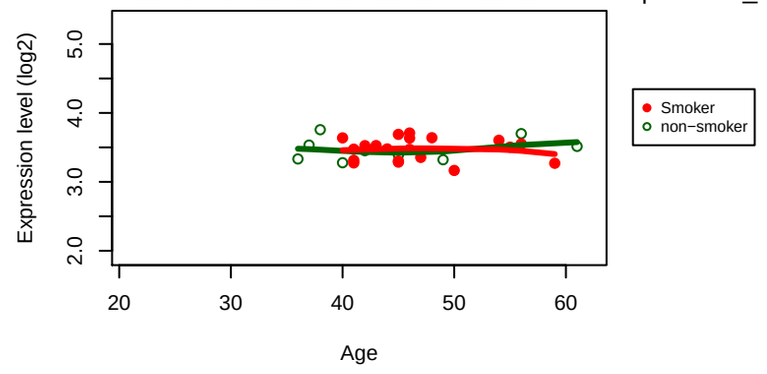
set: SmallAirway1 (p=0.0131)



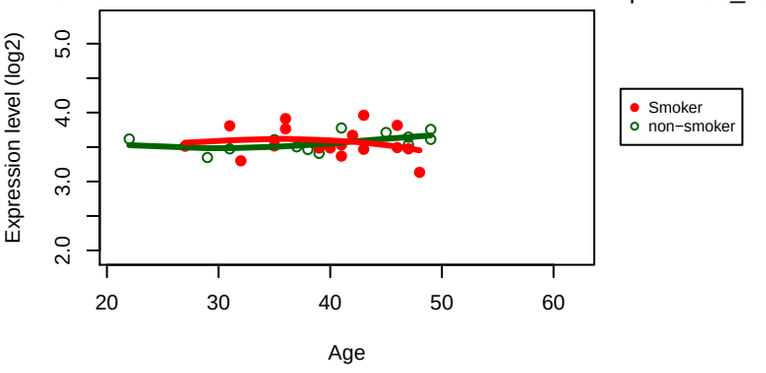
set: SmallAirway2 (p=0.2846)



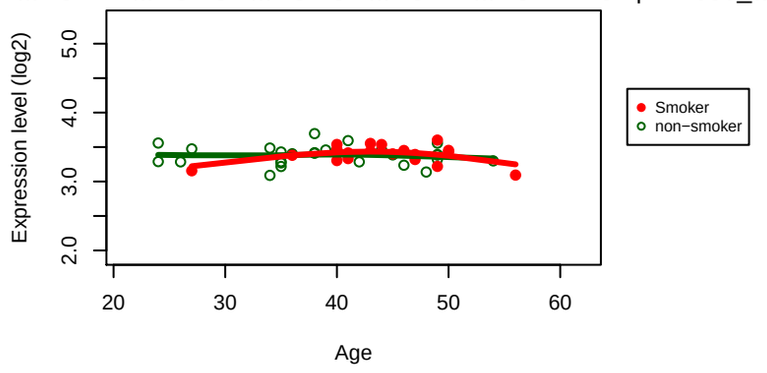
set: LargeAirway1 (p=0.9417)



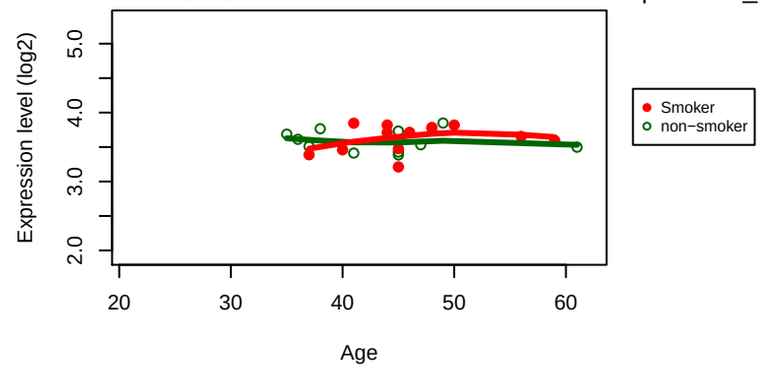
set: LargeAirway2 (p=0.98)



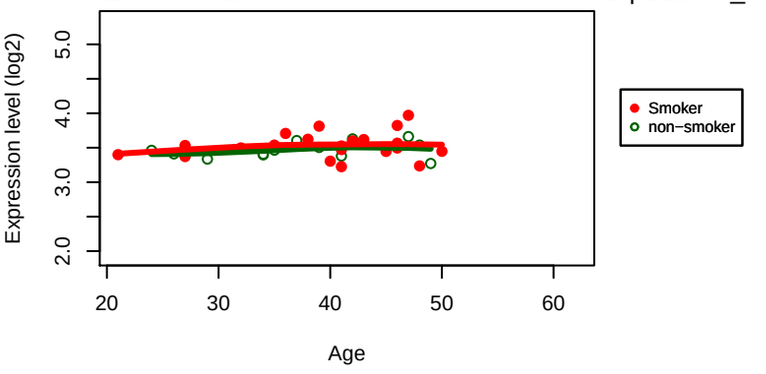
set: Trachea (p=0.6971)

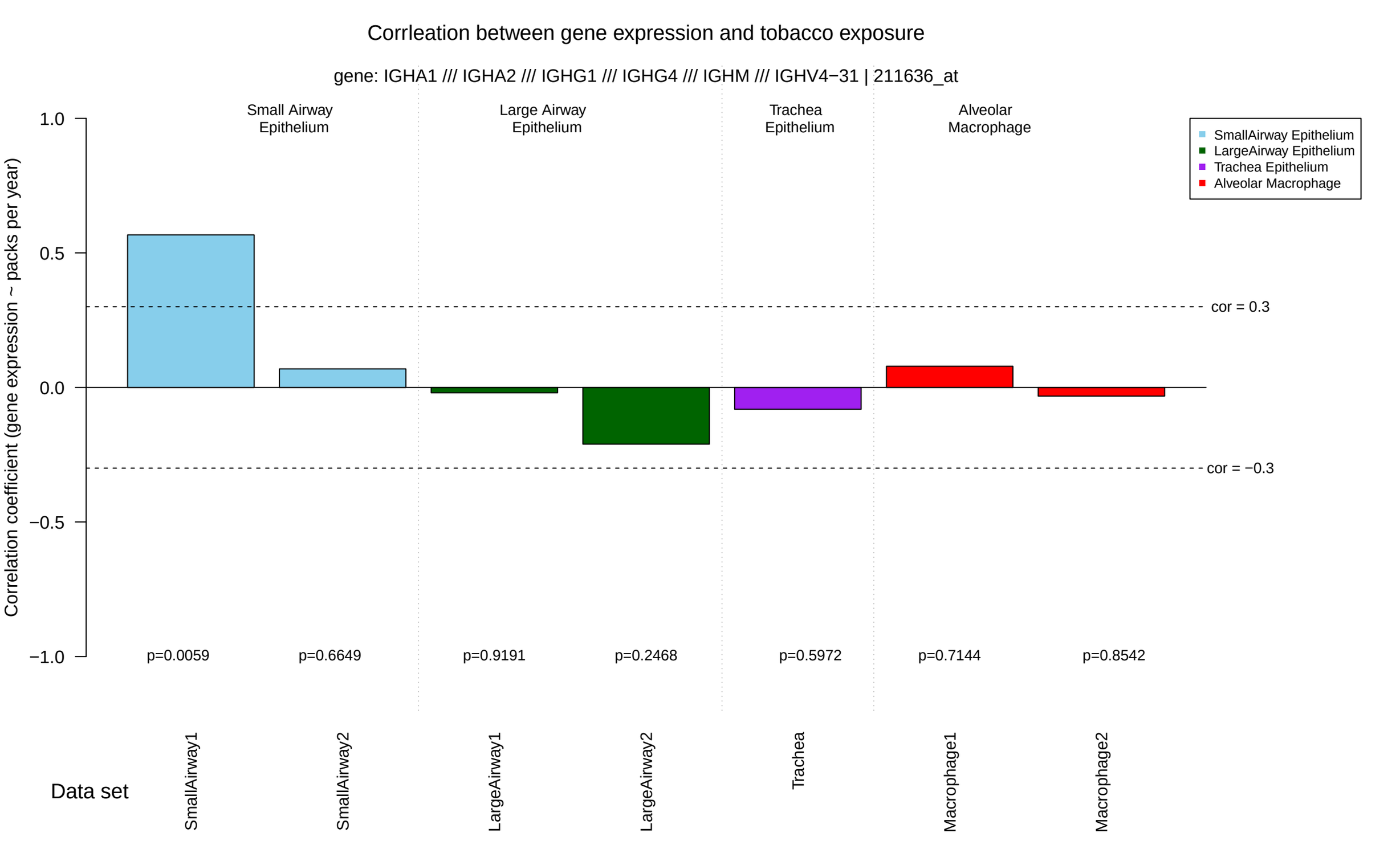


set: Macrophage1 (p=0.479)



set: Macrophage2 (p=0.1983)




Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHA2 /// IGHG1 /// IGHG4 /// IGHM /// IGHV4−31 | 211636_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0059
p=0.6649
p=0.9191
p=0.2468
p=0.5972
p=0.7144
p=0.8542
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2